

# KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

[arxiv.org/abs/2504.03553](https://arxiv.org/abs/2504.03553)

# Current Agent Training Uses 'Flood Irrigation'

## Trajectory Imitation

*(e.g., ReAct)*

Blindly imitates gold trajectories

## Trial-and-Error

*(e.g., Reflexion)*

Refines via external feedback without situational awareness

## Knowledge-Augmented

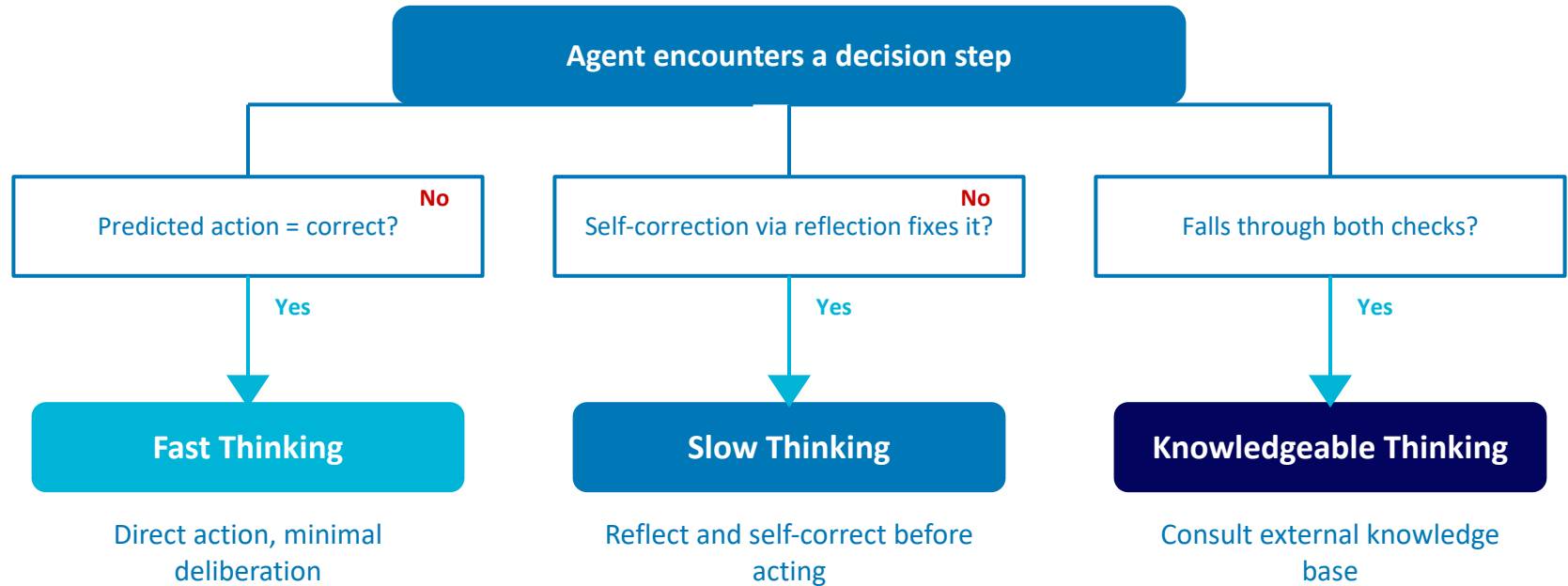
*(e.g., KnowAgent, WKM)*

Injects domain knowledge at every step regardless of need

All three approaches indiscriminately inject information — like 'flood irrigation' — leading to fragile planning, pattern collapse, and high inference costs.

*Humans possess situational self-awareness: knowing when to act, reflect, or seek help.*

# Three Situational Thinking Modes



# KnowSelf Framework: Three-Stage Pipeline

**01**

## Self-Awareness Data Construction

Agent self-explores environments. Heuristic criterion classifies each step: predicted action matches gold → fast; reflection fixes it → slow; otherwise → knowledgeable. Special tokens inserted.



**02**

## Two-Stage Training

SFT teaches initial self-awareness patterns. RPO (Reward-based Policy Optimization) refines special token generation.



**03**

## Self-Aware Inference

Agent generates special tokens that trigger: direct action, reflection, or knowledge retrieval from external KB.

# Qualitative Examples: Where Baselines Fail, KnowSelf Succeeds

Task: Put a hot mug in cabinet

**OpenAI-O1**



Incorrectly tries to take an egg from the microwave after opening it

**KnowSelf (Llama-8B)**



Uses Reflection token to realize it should heat the mug using the open microwave

Task: Put a saltshaker in drawer

**DeepSeek-R1**



Reflects that drawer is open and tries to close it first (fails)

**KnowSelf**

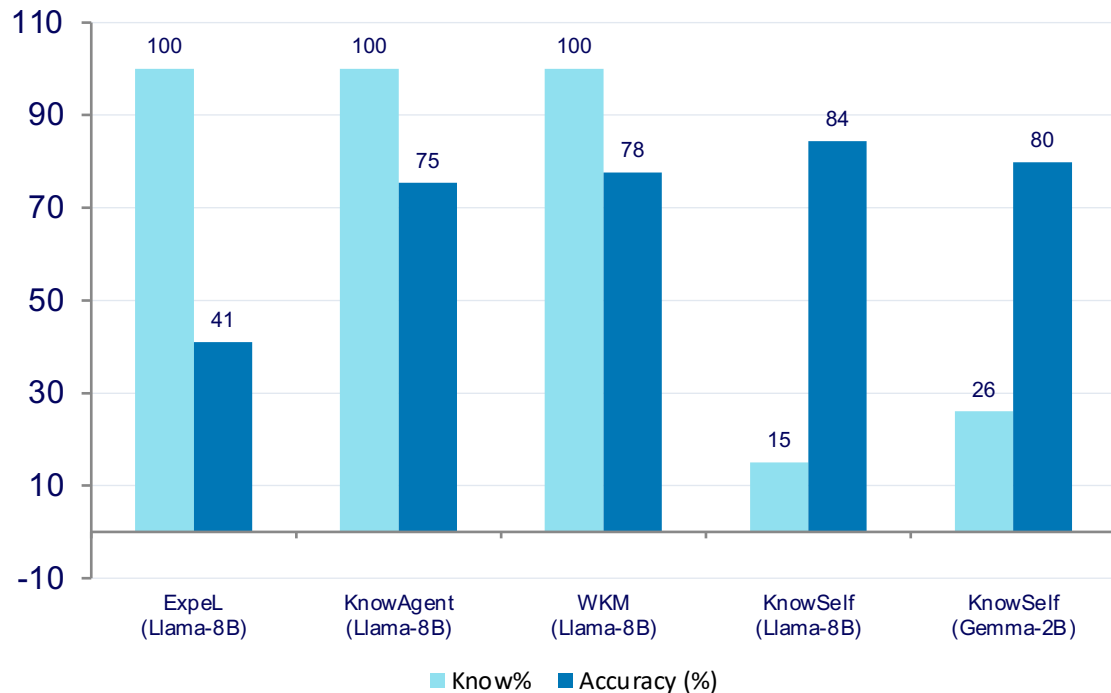


Applies Knowledge token: place the object without removing unrelated items

# ALFWorld Results: Success Rates (%) by Task Type

| Backbone        | Method          | Know%         | Put          | Clean        | Heat         | Cool         | Examine      | Put Two      | All          |
|-----------------|-----------------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| GPT-4o          | ReAct           | 0%            | 83.33        | 74.19        | 69.57        | 85.71        | 77.78        | 64.71        | 76.12        |
| GPT-4o          | Reflexion       | 0%            | 100.00       | 87.10        | 73.91        | 90.48        | 83.33        | 70.59        | 85.07        |
| GPT-4o          | ExpeL           | 100%          | 95.83        | 83.87        | 69.57        | 80.95        | 88.89        | 52.94        | 79.85        |
| Llama-8B        | ReAct           | 0%            | 33.33        | 3.23         | 0.00         | 57.14        | 66.67        | 23.53        | 27.61        |
| Llama-8B        | Reflexion       | 0%            | 62.96        | 22.73        | 5.88         | 64.29        | 86.36        | 50.00        | 51.49        |
| Llama-8B        | ExpeL           | 100%          | 83.33        | 32.26        | 30.43        | 23.81        | 55.56        | 17.65        | 41.04        |
| Llama-8B        | ETO             | 0%            | 91.67        | 70.59        | 82.61        | 61.90        | 88.89        | 64.71        | 78.36        |
| Llama-8B        | KnowAgent       | 100%          | 87.50        | 93.55        | 65.22        | 66.67        | 61.11        | 64.71        | 75.37        |
| Llama-8B        | WKM             | 100%          | 95.83        | 87.10        | 86.96        | 61.90        | 66.67        | 52.94        | 77.61        |
| <b>Llama-8B</b> | <b>KnowSelf</b> | <b>15.01%</b> | <b>91.67</b> | <b>87.10</b> | <b>91.30</b> | <b>85.71</b> | <b>77.78</b> | <b>64.71</b> | <b>84.33</b> |
| Gemma-2B        | ReAct           | 0%            | 0.00         | 9.68         | 0.00         | 4.76         | 44.44        | 0.00         | 8.96         |
| Gemma-2B        | ETO             | 0%            | 91.67        | 83.87        | 78.26        | 52.38        | 77.78        | 29.41        | 71.64        |
| Gemma-2B        | KnowAgent       | 100%          | 91.67        | 90.32        | 69.57        | 71.43        | 66.67        | 41.18        | 73.88        |
| Gemma-2B        | WKM             | 100%          | 91.67        | 87.10        | 78.26        | 71.43        | 61.11        | 52.94        | 76.12        |
| <b>Gemma-2B</b> | <b>KnowSelf</b> | <b>26.41%</b> | <b>87.50</b> | <b>93.55</b> | <b>73.91</b> | <b>76.19</b> | <b>83.33</b> | <b>52.94</b> | <b>79.85</b> |

# Minimal Knowledge, Maximum Impact



KnowSelf uses only 15–26% knowledge queries vs 100% for competitors

Yet achieves the highest accuracy: **84.33%** (Llama-8B)

Scales down: Gemma-2B reaches 79.85%, matching GPT-4o Expel

# Key Takeaways

- Indiscriminate knowledge injection is wasteful and often counterproductive — methods using 100% knowledge frequently underperform KnowSelf

- Agents can learn situational self-awareness through lightweight special-token mechanisms with heuristic labeling — no expensive human annotation needed

- KnowSelf (Llama-8B) achieves 84.33% on ALFWorld, surpassing GPT-4o ExpeL (79.85%) while using only 15% external knowledge

- The three-situation taxonomy (fast/slow/knowledgeable thinking) provides a principled metacognitive framework for autonomous cognitive resource regulation

- The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL

*Future Directions: Extending to multi-agent settings, more complex environments, and integrating with reasoning-augmented LLMs*